

Lab 26 Worksheet — Mobile Device Hardware Fault Diagnosis

Name: ____ Date: ____

Exam objective: Diagnose mobile device hardware faults including battery, charging, screen, digitizer, port and overheating problems, applying the correct safety response (Core 1 objective 5.5).

Record as you go

Step	What you did	What you observed
1	Record the safety rule that governs this entire lab: a swollen battery must never be punctured, compressed or charged. Power the device down, isolate it from flammable material and follow the organisation's hazardous disposal procedure.	
2	Check battery health from the device's own settings, recording the maximum capacity percentage and the cycle count where the operating system exposes them.	
3	Diagnose poor battery life by separating its causes: a genuinely degraded battery, background applications, a weak cellular signal forcing the radio to full transmit power, or high screen brightness.	
4	Diagnose a swollen battery from its physical signs — a lifting screen, a bulging back cover or a device that no longer sits flat — and apply the safety response immediately.	
5	Diagnose improper or intermittent charging by testing in a fixed order: substitute the cable, substitute the charger, inspect the port for lint and damage, then suspect the battery.	
6	Record that a charging port packed with pocket lint is extremely common and is cleared with a	

Step	What you did	What you observed
	wooden or plastic pick on a powered-off device — never a metal tool.	
7	Diagnose a broken screen by distinguishing cracked glass with a working display, a working digitizer with a dead display, and a working display with a dead digitizer.	
8	Diagnose digitizer failure as touch that is unresponsive, offset from where you press, or registering phantom touches, while the image remains perfect.	
9	Diagnose overheating from its causes: sustained high load, a faulty charging circuit, an ambient heat source, or a failing battery — and record that persistent overheating with a swollen battery is an immediate safety stop.	
10	Diagnose liquid damage, record that the device must be powered off and not charged, and note that visible corrosion means board-level damage requiring specialist repair.	
11	Diagnose physically damaged ports and record the repair options in order of cost: port replacement by a specialist, board replacement, or device replacement.	
12	Complete the map with poor or no connectivity, cursor drift requiring touch recalibration, and malware symptoms, giving a technical and a safety response for each.	

Verification

Success criterion: Every fault in your map has both a technical and a safety response; the swollen battery entry forbids puncturing, compressing and charging; and the charging diagnosis substitutes cable and charger before the battery is suspected.

- I completed every step in the lab.
- My result meets the success criterion above.

- I recorded my evidence (screenshots, output, completed tables).

Reflection

What surprised you in this lab?

Where would you apply this on the job?

What do you still need to revise before the exam?
